

Inductors. At high frequencies, inductors behave as series resistors and parallel capacitors besides their natural inductance.

Capacitors. At high frequencies, capacitors behave as series resistors and series inductors besides their natural capacitance.

Thus, simple voltage/current relationships for ideal components are no longer valid at high frequencies and suitable circuit analysis methods are to be used by replacing them with their high-frequency equivalent circuits.

PCB tracks

PCB tracks behave like wires at low frequencies. This behaviour is no longer sustained at high frequencies. To understand this behaviour, you have to understand how a signal is transmitted from a source to the receiver and transmission line theory. Fig. 8 shows a source/driver delivering signal to a load/receiver through wires (shown by lines).

At low frequencies, connecting wires from the transmitter/source to the receiver/load have very low resistance and act as short-circuit. Voltage from the source is directly applied to the receiver. Any change in voltage at the source is delivered instantaneously at the same time to the receiver. Appropriate current flows from the positive lead of the source through connecting wire and the load and the return wire, to the negative terminal of the source. The value of current depends on the source voltage and the load.

Normally, source and receivers are components soldered on PCBs and connecting wires are tracks on PCBs. Send and return PCB tracks behave like a transmission line if the signal frequency of the source is sufficiently high.

Lumped transmission line model represents send and return lines to have resistance (R), inductance (L), capacitance (C) and conductance (G) per unit length. Their values depend on spacing between send and return lines, thickness of lines, transmission medium, etc. Since these lines are copper tracks on PCBs, their



Fig. 4: Equivalent circuit for a wire at high frequency

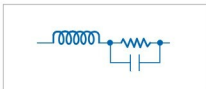


Fig. 5: Equivalent circuit for a resistor at high frequency

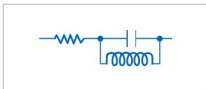


Fig. 6: Equivalent circuit for an inductor at high frequency

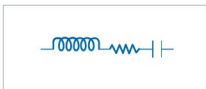


Fig. 7: Equivalent circuit for a capacitor at high frequency

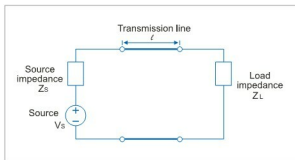


Fig. 8: PCB tracks

values are dependent on PCB tracks' thickness and separation, that is, PCB tracks' geometry and board material. These parameters cause voltages and currents to travel like waves through transmission lines (PCB tracks).

Any change in voltage/current at the source is transmitted like a wave and reaches the receiver after wave propagation delay. This delay is dependent on transmission line

parameters, that is, PCB tracks geometry and board material. At low frequencies of source signal, this delay is much smaller than rise/fall time of the signal. At sufficient high clock/signal edge rates, delay becomes comparable to the rise/fall time of the source signal/clock.

Transmission lines have an important characteristic called characteristic impedance (Z_0), which depends on R, L, G, C and thus PCB tracks geometry and board material. Equations derived in transmission line theory using lumped parameter model show that at any instant of time, and at any point on transmission line, voltage/current values are obtained as the sum of a forward travelling wave from the source to the load and a reverse travelling wave from the load to the source. The reverse travelling wave is called reflected wave from the load.

If load impedance exactly equals Z_0 , amplitude of the reflected wave will be zero and there will be no reflected wave from the load. In this case, signals from the source are transmitted to the load without any distortion.

If load impedance does not equal Z_0 , there will be a reflected wave from the load to the source. This situation is called impedance mismatch. If there exists impedance mismatch at source side also, the reflected wave will be reflected back again from the source to the load. Thus there will be multiple reflections on both sides. Transmission line also attenuates the amplitude of forward and reflected waves along its length. Hence the amplitude of each successive reflected wave is reduced. These impedance mismatches and reflections cause the signal sent from the source to be distorted at the load end, giving rise to a phenomenon called ringing. This can be avoided by following appropriate resistor termination practices at the load side and/or the source side.

The rule of thumb for considering PCB tracks from a source to the load as transmission line is that the total propagation delay of the signal